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Reinforcement Learning-based multimodal traffic signal control considering pedestrians and cyclists in real scenarios

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Reinforcement Learning-based multimodal traffic signal control considering pedestrians and cyclists in real scenarios

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Però tu sabràs el que fas, jo vindria a practicar el noble art d'anar endavant.
Si et sents intrús no parlis gaire i ja veuràs, ningú adverteix l'engany;
no et demanen un carnet, no hi ha escrit cap reglament, s'equivoca el més expert.
Donem conversa, som amables, i no se sap si malgrat l'esforç anem
cap al plaer o al dolor,
a la llum o a la foscor,
a un gran banquet o a la més cruel inanició.

Tots ens movem, no pots quedar-te encallat eternament;
avança, vianant, per les llambordes i l'asfalt!

MANEL – *Avança, vianant*

To the memory of Miquel Àngel Llauger i Lull

1926 – 2009

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Abstract

Reinforcement learning can provide robust algorithms for traffic signal control. However, these approaches usually focus exclusively on automobiles, not accounting for other road users like pedestrians and cyclists. This limitation undermines their applicability to real scenarios and neglects the relevance of active mobility modes in transportation systems. This thesis aims to extend the reinforcement learning framework developed by Rodrigues and Lima Azevedo (2019) to include pedestrian and cyclist agents. The thesis will design realistic multimodal traffic scenarios in SUMO, adapt RL representations to incorporate pedestrians and cyclists, and redefine the reward function to consider all users. The expected outcome is the development of a more realistic framework that enables a greater understanding of urban mobility.

Resumen

El *reinforcement learning* (RL, aprendizaje por refuerzo) puede proporcionar algoritmos robustos para el control de semáforos. Sin embargo, estas aproximaciones suelen centrarse exclusivamente en automóviles, no considerando a otros usuarios como peatones y ciclistas. Esta limitación socava su aplicabilidad a escenarios reales y no tiene en cuenta la relevancia de los modos de movilidad activa en los sistemas de transporte. Esta tesis pretende extender el marco de *reinforcement learning* desarrollado por Rodrigues y Lima Azevedo (2019) para incluir agentes peatonales y ciclistas. El trabajo diseñará escenarios realistas de tráfico multimodal en SUMO, adaptará representaciones de RL para incorporar a peatones y ciclistas, y redefinirá la función de recompensa para considerar a todos los usuarios. El modelo resultante será evaluado en escenarios reales. El objetivo esperado es el desarrollo de un marco más realista que permita una mejor comprensión de la movilidad urbana.

Abstrakt

Forstærkningslæring (RL) kan give robuste algoritmer til trafiksignalstyring. Disse tilgange fokuserer dog normalt udelukkende på biler og tager ikke højde for andre trafikanter som fodgængere og cyklister. Denne begrænsning underminerer deres anvendelighed i virkelige scenarier og negligerer relevansen af aktive mobilitetsformer i transportsystemer. Denne afhandling sigter mod at udvide rammen for forstærkningslæring udviklet af Rodrigues og Lima Azevedo (2019) til at omfatte fodgængere og cyklister. Afhandlingen vil designe realistiske multimodale trafikscenarier i SUMO, tilpasse RL repræsentationer til at inkorporere fodgængere og cyklister og omdefinere belønningsfunktionen til at tage alle brugere i betragtning. Det forventede resultat er udviklingen af en mere realistisk ramme, der muliggør en større forståelse af bymobilitet.

Resum

El *reinforcement learning* (RL, aprenentage per reforç) pot proporcionar algoritmes robusts per el control de semàfors. No obstant, aquestes aproximacions solen centrar-se exclusivament en automòbils, no considerant altres usuaris como vianants i ciclistes. Aquesta limitació menyscava la aplicabilitat a escenaris reales i no té en compte la rellevància dels modes de mobilitat activa als sistemes de transport. Aquesta tesis pretén extendre el marc de *reinforcement learning* desenvolupat per Rodrigues i Lima Azevedo (2019) per a incloure agents de vianants i ciclistes. El treball dissenyarà escenaris realistes de trànsit multimodal a SUMO, adaptarà representacions de RL per a incorporar a vianants i ciclistes, i redefinirà la funció de recompensa per a considerar tots els usuarios. El model resultant serà aval·luat a escenaris reals. L'objectiu esperat és el desenvolupament d'un marc més realista que permita una millor comprensió de la mobilitat urbana.

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Introduction 1

It was the year of 1868 when the first traffic signals in the world came to be, installed outside the Houses of Parliament in London. The manually operated gas-lit signals that the railway engineer J.P. Knight designed were not, however, intended to control an intersection between busy flows of horse-drawn carriages, but rather to allow the safe crossing of a street: the reason to be of the first traffic signal was to serve the pedestrian.

Along with the evolution and expansion of the automobile, traffic signals also evolved and became more ubiquitous in cities around the world. The shiny novel vehicles quickly became traffic, a fast and dangerous new kind of it, and that traffic needed to be controlled; traffic signals —or traffic lights, or *robots*, as known in South Africa— were the logical choice for that.

The high regard for the pedestrian that gave the traffic light its conception was quickly forgotten in the process. As streets and roads surrendered more and more space to the car, more and more engineers and architects preached its new rule as a symbol of progress. For many, the priority was to move cars, not people, and that reflected on traffic signals, as will be seen in the pages that follow.

Despite the shift that mobility has experienced ever since, placing other modes of transport in their rightful place, some of those habits and designs continue to be present in our cities and traffic signals are no exception.

When I moved to València for the start of my civil engineering education, traffic signals quickly became an unexpected topic in which to ponder. While in my native Palma I was used to waiting for the green light before crossing the street, to do so in València felt absurd. When walking in Palma, a red light meant that cars were in the way: waiting for the green one was not a matter of compliance but of self-preservation; in València, a red light as a pedestrian meant that *maybe* some cars could appear, almost always in a rhythm and quantity that did not call for the length of the halt.

It is well known that traffic control devices influence people's behaviour,[1] and unsurprisingly, I frequently saw valencians jaywalking to avoid long waits in front of an empty roadway. This practice was not limited to people in a hurry or to rebellious types: families with children, elderly people, parents with strollers and in general anyone who otherwise would be expected to be a law-abiding[2] citizen would usually just check if traffic was clear and cross, ignoring the red light. The refusal to follow the traffic signals extended beyond pedestrians, as cyclists and motorized scooter users often did the same.

[1] For the case of traffic signals, see for instance tal y cual

[2] Pursuant to article 145 of the Spanish traffic law (*Reglamento General de Circulación* [8]), ignoring a red light as a pedestrian is a «severe» infraction, punishable with a fine of 200 euros. In València, I never saw the police prosecuting anyone for it, even when the jaywalking was done directly in front of the officers.

I observed that, while more widespread in side streets and other lightly trafficked ways, these attitudes were still common even in wide avenues where the crossing involved many lanes of traffic at high speeds; thoroughfares which in other cities would have been perceived as dangerous to cross without clearance.

It felt logical: if during everyday walks or bike rides a person constantly encountered traffic signals ordering them to stop, sometimes reasonably, sometimes for periods double the time that the motorized traffic required, sometimes at places where the actual necessity of the traffic signal itself came into question; a red light would lose its meaning to them, and hence, its use as a means of protecting people.

I unfortunately saw many accidents resulting from pedestrians and cyclists not following traffic signals during my studies in València, most of the time just a fright that left everyone unharmed, but occasionally the consequences were much more severe.

I felt that this situation was not only unfair but dangerous, and I could not

conductor luz verde alguien q no miro bien o mala bisibilidad Luego q digo esto es injusto y also peligroso es q nadie ha programao estos semaforos??? como puedee ser Luego lo de centro de control ironicamente q fui y pregunte, espiras ref

1.1 Motivation

Lorem ipsum

Stopping for a red light is a primary discouragement to cycling ([67]); it costs a great deal of extra time and energy and imposes additional stress on the body which can lead to injury. In addition, according to Sharples, the peak hour for cyclists is much more pronounced than for motorized traffic (Sharples, 1997) Y VER CPH and commuting cyclists prefer controlled environments, for instance traffic controlled intersections (Aultman-Hall, Hall, and Baetz, 1997). Based on these statements, it becomes easily understandable that intervention at signalized intersections in favor of cyclists is more than justified. SIGUE LEYENDO EN [2] para más motivos sobre infraccion de rojos por ciclistas que a su vez se cobra vidas, Y LAS DELAYS DISCOURAGEAN AL CICLISMO

1.2 Problem statement

Lorem ipsum

1.3 Research objectives

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1.4 Scope and limitations

State of the art 2

2.1 Traffic signal control as a multimodal problem

As introduced in the preface, traffic signals were, from their inception, designed to manage multimodal traffic, and not just motor vehicles [7]. However, as motorization increased throughout the 20th century, traffic lights quickly centred around the needs of the automobile; once the first automatically timed traffic signals were installed in the 1920s, their time periods were set using traffic surveys, and the later development of green wave systems continued the trend. [55]

Despite remaining an anecdote, the 1931 letter of the Swedish traffic engineer Perikö af der Stykker to the *American City Magazine* illustrates frankly the treatment of non-motorized users at the time. Af der Stykker, who was the traffic engineer of Stockholm, was planning to add pedestrian signals to the city's new traffic lights. As the latest developments in traffic signals were then occuring in the United States —by then, systems incorporating turn lane phases, management of reversible lanes or traffic activation were already in operation there [55]—, af der Stykker asked his US colleagues if they had tried any similar solution. They had not. [32]

enlazar con lo anterior a nivel network q Vehicle delay is perhaps the most important parameter used by transportation professionals to evaluate the performance of signalized intersections??? o otros parámetros???

Noland has shown that focusing only on motorvehicle flows may not lead to economically optimal solutions whenpedestrian delay is also accounted for ([59]). For some trips, the traveltime delay costs associated with walking may reduce the likelihoodthat people will choose to walk.

diciendo q research focus en coxes, de ahi pasamos a multimodal

Network optimisation has traditionally been approached as a problem of vehicle flow optimisation. The various approaches developed for optimisation of signal-controlled intersections are no exception. Webster's ([76]) and other numerous methods that have since been formulated for optimisation of signalised intersections perform stage- or phase-based optimisation for conflicting streams of vehicle flows but assign fixed times to the pedestrian phase exogenously, not taking into account pedestrian flows and delays.

Only a few research studies have considered pedestrian travel cost as a constituent of the criteria for design of traffic control elements. An example is Griffiths et al. (1976) who based the selection of type of control (zebra or pelican) at pedestrian crossings on comparative values of time of different modes. Disregarding pedestrian travel costs might lead to non-optimal network design if pedestrians form a noticeable proportion of road users, such as in urban centres

q mayoria trabajos que consideran peats pero en intersecciones concretas, no a nivel de red, y que no consideran ciclistas ni tampoco la variabilidad de los peats, y que además no usan RL sino optimización tradicional, o fuzzy logic (justificar xq eso no sirve para lo nuestor)

Previous works have been done to dynamically control adaptive traffic lights. But due to the limited computing power and simulation tools, early studies focus on solving the problem by fuzzy logic [19], linear programming [20], etc. In these works, road traffic is modeled by limited information, which cannot be applied in large scale.

OBJS NO TRADICIONALES: vilarinho y also incorpora peds [72]

Li et al. [41] proposed an optimization model to improve vehicle and pedestrian efficiency, the results have shown that considering pedestrian and vehicle delays together improves the model performance. Other studies [88] [21], [22] also sought to minimize a combination of vehicle and pedestrian delays. Beyond optimizing traffic efficiency, traffic safety also attract researchers' attention. Ma et al. [23] minimized total monetary costs that include traffic delays as well as potential pedestrian/vehicle conflicts. Considering the allotted pedestrian clearance time may fail to clear the crosswalk, which will increase the probability of pedestrian-vehicle exposures, Zhang et al. [24] introduced an additional dynamic all red phase to balance traffic efficiency and safety also single one pero famoso [42]

SUMO [1]

Ishaque, et al. ([33]; [34] el segundo accounts for non compliance) analyzed the trade-offs in pedestrian and vehicle delays in a hypothetical network by considering relative values of time for pedestrians and vehicles. They found that shorter cycle lengths are beneficial for pedestrians. Moreover, the existing policies that are most advantageous to vehicles might be disadvantageous to pedestrians, which do not make the network optimally perform for all road users. Although, they assumed that pedestrian delay is composed only from control delay. Actually with high demands, pedestrians experience significant delays while discharging at the edge of the crosswalk and while crossing the street due to the interaction between opposing pedestrian flows. Furthermore, a discussion about the optimized signal parameters considering pedestrian and vehicle delays was not presented. Few studies addressed the issue of bi-directional pedestrian flow and its impact on crossing time and speed at signalized crosswalks and the resultant delays. HCM does not consider the effects of pedestrian demand and crosswalk width on pedestrian crossing time. However when pedestrian demand increases at both sides of the crosswalk, crossing time increases due to the interaction between conflicting pedestrian flows. Algo similar a Ishaque usando el software Synchro hicieron en [6]

[88] usa meta-heuristics con discrete harmony search (DHS) algorithm en red en cuadrícula considerando peds

The artificial intelligence methods have also been applied on adaptive traffic signal control. Signal controllers using reinforcement learning approaches are developed via the networkwide multi-agent coordination and the relationship between actions, states and environment [24], [25]. Although no prespecified traffic model is required, the exponential growth of the state-action space with the increase of the agents may fail to efficiently solve the problem. Neural network (NN) is another technique that has been employed for signal controllers. A fuzzy NN system is formulated via the simultaneous perturbation stochastic approximation (SPSA) algorithm to update the weight online [26]. Chao et al. [27] proposed an intelligent traffic light control strategy based on extension neural network theory (ENN) for isolated intersection. However, NN-based methods need to consider appropriate number of layers and neurons, also, they are not easily adjustable when new traffic situation occurs. Moreover, fuzzy logic (FL) is adopted in traffic signal control to mimic human thinking or experts' knowledge when the mathematical model is very difficult to formulate [28], [29]. The cellular automata (CA) mechanism with particle swarm optimization method is proposed to solve traffic light scheduling problem [30]. Also, a CA-based multi-intersection model representing the evolution of the traffic network is developed to reduce the traffic pressure [31]

Review of lit para mixed traffic flows: 2023, [52] mencionar de pasada y au xq mixed es vehiculos autonomos, verifica q no haya nada con peds tho

qlearning algorythm en network [47]

pedestrians con camaras nsk creo q no es RL [66]

[2] During the mid 1980's, in Beijing, in China, the increased traffic volumes lead the authorities to implement an urban traffic and control system in a section of the city. The 11 CHAPTER 2. STATE-OF-THE-ART chosen system was the SCOOT traffic control system (Split Cycle Offset Optimization Technique). The system takes into account the cyclists traveling along a link, by detecting them with a loop detector, which is located a short distance from the stop line. The team established a composite measure of bicycle flow and occupancy called a Bicycle Link Profile Unit (BLPU) which was capable of aggregation with the normal SCOOT Link Profile Unit (LPU). This enabled the bicycle link to be incorporated into the standard SCOOT model in the same manner as other links. Weighting functions are then available to advantage or disadvantage the bicycle traffic stream relative to the motorized traffic stream

A European example is the historic city of York in the UK. Within York the level of commuting by cycles is high at 19deployment of much urban traffic control infrastructure. The local authority has a package of techniques and expertise which it can use in the appropriate circumstances. One techniques is "tailend biasing" where the offsets at the ends of greens are co-ordinated to provide for slower modes such as buses and cyclists. Another technique is the use of SCOOT detectors in cycle loops and "dummy" SCOOT loops which shadow existing SCOOT links. Increasingly use is made of microwave detectors mounted on signal heads which are able to accurately detect cyclists if they are travelling at 10km/h (6.25 mph) or more.

Green waves, also en apostola: The world's first green wave for cyclists was created in Odense, in Denmark. Along a bicycle route 45 low poles have been placed between two intersections, over a distance of some 350 meters, each provided with a light. The lights turn green one after the other, at an interval of some seconds. Following the timing of the lights, the cyclist has the right speed for a green wave. However, the problem is that cyclists have widely varying cruising speeds (Andersen, 2013). In the autumn of 2006 a green wave for cyclists has been realized on Norrebrogade, in Copenhagen, by the local authorities. The green wave affects 12 traffic lights over a stretch of road of over 2 km. The idea is that if the bicyclist rides 20 km/h, he will hit green lights the whole way into the city center. The traffic lights are coordinated during the morning rush hour between $6:00$ - $10:00/12:00$ (depending on the route) and then the wave reverses so the lights are coordinated in the opposite direction between $12:00/15:00$ - $18:00$ for the trip home. Norrebrogade is a major route for both motorized and bicycle 13 CHAPTER 2. STATE-OF-THE-ART traffic. The results for bicycles are very positive, considering the decrease of number of stops from 6 to 0 and the increase of speed from 15.1 to 20.7 km/h. The green wave for cyclists has also realized on three other roads (Hoegh, 2007). luego de esto muchas otras ciudades

Nondominated Sorting Genetic Algorithm II (NO ACCESS tho) for ebikes worked well [24]

2.2 Reinforcement Learning for traffic signal control

fonament de multiagent RL en TSC: [5]

review lit 2021 [77]

RL solutions WITH PEDS constrained to intersections: [74] (considering large intersections with pedestrian two-stage crosswalks)

[26] muy citao, una sola intersecc y coches, sumo tho

solo una interesecc con peds [80] [92] [29]

solo una intersecc pero primero q considera loc de peds, also peque review de research con peds so far [27]

solo una interescc [10], [57], [70], [27]

solo con una intesecc pero SUMO y CICLISTAS, deep reinf learning [50]

solo una intersecc pero con peds, sumo y vehs autonomos [86]

2 ints solo coches pero CNN+DRL [15]

idem pero con PEDS [85], LEER BIEN Y MIRAR CITADOS y QUIENES LO CITAN

BUENA REVIEW DE LA LIT DE RL AS OF 2024: https://academic.oup.com/iti/article/doi/10 [81]

[3] paper de resco también tiene una lista de metodos de ML aplicados a TSC, incluyendo RL específicamente

Network level, monaco y grid, also interesante [11]

Network level e interesante: [87] (en vd es de 2021)

network level sin peds [49], [13]

grids sin peds pero considerando fairness para patas coches [44] idem pero citao por tener graph nsk [69] idem pero buses no fairness tho [43]

network sencillita con peds [68]

corredor coches y buses no peds [84] y [62], solo coches [25]

idem china [46], also [14] y [45]

also networkk overcomeando movidas [12]

corredor no peds [91]

Network en seul y oversaturated conditions [40]

algo alternativo a RL??? [79]

network en grid [90], idem y tmb es framework [35]

network xikita si considera peats pero con green duration nsk; also itneresante en RL [83]

network siux falls con peds [82]

network con peds [47]

grids tochas no peds [89]

Network no peds [58]

corredor con demanda ped sintetica [37], leer bien y mirar citados y quienes lo citan

desarrolla mas [36], idem

corredor con DEMANDA PED REAL!!! [64], LEER BIEN Y MIRAR CITADOS y QUIENES LO CITAN

REVIEW DE LIT EN 2025: [51]

[75] Grid en beijing, xikita, Traditional methods cannot effectively coordinate multi-intersections with different signal timing time. This has led to the introduction of a novel clock consistency mechanism the operation of which is implemented through a Multi-Agent Coordinated Reinforcement Learning (MACRL) algorithm.

[31] y desarrollado en [30], networks default, Despite its potential, the real-world application of RL-based controllers is constrained by low sample efficiency and high computational demands. To address these challenges, we propose DTLight, a simple yet

powerful lightweight Decision Transformer (DT)-based offline-to-online TSC method that can learn policy from pre-collected offline datasets while maintaining the capability to refine policy with minimal online interactions.

[56] una sola intersecc pero muy interesante, a la vanguardia de peds rn Y ESTAO DEL ARTEEEE as of 2026666

Dice el perplejo: Multimodal RL signal control in SUMO is mostly restricted to single-intersection case studies, with incomplete treatment of non-motorized users and typically without cyclists as a separate mode. To the best of our knowledge, network-level multimodal RL TSC in SUMO with simultaneous motor vehicle, bicycle, and pedestrian flows calibrated from real data has not yet been demonstrated; the closest work (e.g., the TKDE 2025 article you cite) does not include cyclists explicitly.

Ojo la ped safety: los tiempos han de permitir cruzarrr

comparacion de metodos en sumo: [48]

2.3 Modelling Pedestrians and Cyclists in Microscopic Traffic Simulation

Traffic control systems are the most visible element of the urban infrastructure. They are not just physical systems like telephones or sewers or streets, although their technological elements, traffic lights, signs, and painted pavements, fit that description. Rather, they are systems that attempt to impose a strong social control over the most fundamental of human behaviors, whether to move or be still. Traffic engineers must control police, drivers, and pedestrians. For most other elements of the urban infrastructure, controlling the behavior of users did not constitute the primary goal of designers. [55] For traffic engineers, understanding and manipulating the behavioral patterns of drivers and pedestrians (a group that included not just walkers, but people using the street for play, social gatherings, and commerce) proved to be a more important problem than the control mechanisms themselves. Traffic engineers learned rapidly to pay careful attention to ergonomics, the interface between people and machines. There was also a political agenda of maximizing expert control. [55]

Pedestrian crossing speed will directly impact the determination of TSC parameters, such as pedestrian clearance time. Pedestrian crossing speed is influenced by various factors, including age, health condition, physical fitness, and psychological expectations. As a result, there are significant differences in crossing speeds among different individuals. Relevant studies [9] [28] indicate that pedestrian crossing speed at intersections typically follows a normal distribution. However, due to the lack of detailed consideration of pedestrian speed characteristics, most studies adopt fixed clearance time, which may result in potential traffic risks, as illustrated in Fig. 2.

More recent research, conducted through child and adult focus groups by the Transport Research Laboratory, has shown that 30 s is the maximum amount of time that both children and adults are willing to wait at a signalised intersection ([54]). The German Highway Capacity Manual specifically recommends that signal cycle times above 90 s should be avoided to reduce pedestrian delay (Brilon, 1994). In many cases where infrastructure has been designed based on the Transportation Research Board's 'Highway Capacity Manual' (2000) or the UK Highways Agency's 'Design Manual for Roads and Bridges' (Highways Agency, 2005) and related policy guidelines, longer signal cycles are in effect

At present, HCM's (2010) pedestrian delay model is the most commonly used model to estimate pedestrian delay value at signalized intersections. The model is developed considering assumptions such, as uniform arrival rate of pedestrian, pedestrian comply with traffic signal, and fixed cycle length with the delay being dependent only on signal time. It is to be noted that these assumptions are not suitable for mixed traffic conditions prevailing on Indian (DANISH too) roads. [53] proposes models to estimate pedestrian delay considering compliance to signals.

[41] tiene modelos para ped delays tanto en acera como cruzando

q bicis son heterogeneas (como demuestra [65]) y sumo no considera por defecto

han hecho para sumo el Realistic Bicycle Dynamics Model ver si usaremos al final [61], viene de simra creo [38]

[39] framework para modelar bicis en sumo, no usamos xq coste computacional y no impacta en tiempos parece ser segun autores???

2.4 Simulation Environments for RL-Based TSC

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Frameworks en SUMO ya hechos: https://github.com/flow-project/flow tocho pero creo que sin ciclistas ni peds, [78] [73] idem sin ciclistas ni peds pero buena review de research, usa datos de camaras: [4]

RESCO, Andrew parte de él [3], SUMO-based benchmark suite (single and multi-agent) with realistic, calibrated scenarios and standardized RL implementations; traffic is essentially motor vehicles only. This is the canonical reference for RL + SUMO benchmarks and multi-intersection networks. (https://github.com/Pi-Star-Lab/RESCO)

https://github.com/LucasAlegre/sumo-rl ahí hay lista lista de papers basados en SUMO-RL (RESCO entre ellos)

`https://github.com/EEEEEclectic/traffic_signal_control` -, sin cita preparada idk, We integrate approaches like DCRNN and transformer-based models within the SUMO-RL environment and focus on capturing spatio-temporal dependencies in the traffic network via graph structures.

network en grid en SUMO, tmb es framework [35]

Methodology 3

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3.1 Simulation environment

Lorem ipsum hablar d sumo etcs

q usamos jupedsim o want to, el collision free speed model [71] q queremos usar el Realistic Bycicle Dynamics Model [61]

3.2 Scenario design

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3.2.1 Demand generation

3.2.1.1 Data sources

Traffic count data was generated by the rom the COMPASS system [63], explicar columnas etc, corresponding to *Annual Average Daily Traffic* (AADT) etc. The data generated corresponds to the model's base calculation for the year 2025.

The AADT count data was converted to peak hour traffic volumes using a peak hour k factor so that

$$\text{Peak hour volume} = k \cdot \text{AADT} \tag{3.1}$$

The *Federal Highway Administration of the United States* (FHWA) states that in urban areas the k value is placed towards the lower end of the $0.07 - 0.12$ range [60]. To obtain a better estimate of the factor, the AADT values for several locations obtained with COMPASS were assessed against the peak hours observed from actual count data for the same locations. A k factor of 0.075 was found to provide close enough estimates for the different modes.[1]

3.3 Reinforcement learning framework

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3.4 Definitions and architecture

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[1]The counts assessed correspond to the ones reported in [23], [21], [22], [16], [20], [18], [17] and [19].

Evaluation and results 4

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4.1 Evaluation metrics

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4.2 Results

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Discussion 5

Lorem ipsum

5.1 Interpretation of results

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5.2 Implications for urban mobility

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5.3 Limitations of the study

Lorem ipsum Lorem ipsum V2Infra comms con vehiculos coebcatdos, research q hay etc

Conclusions 6

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6.1 Conclusions

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6.2 Future work

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Appendix A

Appendix 2.1: Chapter 2 - Case study model parameterizations and procedure

Expanded Food Web

The model is of the form:

$$\frac{dN_x}{dt} = r_x N_x - m_x N_x^2 + \sum_{y \in S, y \neq x} a_{xy} N_x N_y \tag{A.1}$$

where S is the set of species. Non-trophic interactions (NTI) may affect the growth rate r_x, mortality rate m_x or interaction strength a_{xy} terms. The ones included in our model are, respectively:

growth rates

$$r_{CM} = \frac{r_{CM}^{NTI} N_{PL} + r_{CM}^{0} N_{PL}^{0}}{N_{PL} + N_{PL}^{0}} \tag{A.2}$$

$$r_{DP} = \frac{r_{DP}^{NTI} N_{HM} + r_{DP}^{0} N_{HM}^{0}}{N_{HM} + N_{HM}^{0}} \tag{A.3}$$

$$r_{PL} = \frac{r_{PL}^{NTI} (N_{CM} + N_{HM} + N_{PLe}) + r_{PL}^{0} (N_{CM}^{0} + N_{HM}^{0} + N_{PLe}^{0})}{(N_{CM} + N_{HM} + N_{PLe}) + (N_{CM}^{0} + N_{HM}^{0} + N_{PLe}^{0})} \tag{A.4}$$

$$r_{PLe} = \frac{r_{PLe}^{NTI} N_{PL} + r_{PLe}^{0} N_{PL}^{0}}{N_{PL} + N_{PL}^{0}} \tag{A.5}$$

mortality rates

$$m_{HM} = \frac{m_{HM}^{NTI} N_{SV} + m_{HM}^{0} N_{SV}^{0}}{N_{SV} + N_{SV}^{0}} \tag{A.6}$$

interaction terms

$$a_{PL,DP} = \frac{a^{NTI}_{PL,DP} N_{HM} + a^{0}_{PL,DP} N^{0}_{HM}}{N_{HM} + N^{0}_{HM}} \tag{A.7}$$

$$a_{DP,PL} = \frac{a^{NTI}_{DP,PL} N_{HM} + a^{0}_{DP,PL} N^{0}_{HM}}{N_{HM} + N^{0}_{HM}} \tag{A.8}$$

We performed two sets of simulations, with and without non-trophic interactions (NTI henceforth). In the simulation without NTI, we assumed that *Podarcis lilfordi* (PL) consumed seeds of *Helicodiceros muscivorus* (HM), *Pistacia lentiscus* (Ple) and *Chritmum maritimum* (CM); these interactions were modelled as mutualisms in the NTI simulation, with the presence of PL individuals increasing the growth rate of each associated plant species. Each simulation was replicated 100000 times for 2500 timesteps. In the main text we report the aggregated results of the 100000 replicates. Each parameter was assigned a minimum and maximum value, and in each replicate parameter values were taken randomly from these intervals.

Parameter ranges for the simulation without NTIs:

Growth and mortality rates:

$$\begin{aligned}
r_{FT} &= [-0.01, -0.001]\\
r_{PL} &= [0.01, 0.1]\\
r_{DP} &= [0.4, 0.6]\\
r_{SV} &= [0.05, 0.15]\\
r_{HM} &= [0.15, 0.25]\\
r_{PLe} &= [0.05, 0.15]\\
r_{CM} &= [0.15, 0.25]\\
m_{FT} &= [0.0001, 0.0015]\\
m_{PL} &= [0.0001, 0.0015]\\
m_{DP} &= [0.0005, 0.0015]\\
m_{SV} &= [0.0005, 0.0015]\\
m_{HM} &= [0.0005, 0.0015]\\
m_{PLe} &= [0.0002, 0.0004]\\
m_{CM} &= [0.00005, 0.00015]
\end{aligned}$$

Interaction coefficients $\neq 0$:

$a_{FT,PL} = [10^{-6}, 10^{-4}]$
$a_{PL,FT} = [-10^{-2}, -10^{-4}]$
$a_{PL,DP} = [10^{-4}, 10^{-2}]$
$a_{PL,HM} = [10^{-6}, 10^{-4}]$
$a_{PL,PLe} = [10^{-7}, 10^{-5}]$
$a_{PL,CM} = [10^{-7}, 10^{-5}]$
$a_{DP,PL} = [-10^{-5}, -10^{-7}]$
$a_{DP,HM} = [10^{-6}, 10^{-4}]$
$a_{HM,PL} = [-10^{-6}, -10^{-8}]$
$a_{HM,DP} = [-10^{-6}, -10^{-8}]$
$a_{PLe,PL} = [-10^{-6}, -10^{-8}]$
$a_{CM,PL} = [-10^{-6}, -10^{-8}]$

Initial abundances:

$N_{FT} = 8$
$N_{PL} = 5000$
$N_{DP} = 200$
$N_{SV} = 5000$
$N_{HM} = 5000$
$N_{PLe} = 200$
$N_{CM} = 5000$

Parameter ranges for the simulation with NTIs. Parameters without superscript are not affected by NTIs. Parameters with superscript 0 indicate values in the absence of NTI, i.e. when one of the interacting species is not present. Parameters with superscript NTI indicate the maximum value that the parameter can reach with NTI:

$$
\begin{aligned}
&r_{FT} = [-0.01, -0.001]\\
&r^0_{PL} = [0.01, 0.1]\\
&r^{NTI}_{PL} = [0.15, 0.25]\\
&r^0_{DP} = [0.65, 0.75]\\
&r^{NTI}_{DP} = [0.4, 0.6]\\
&r_{SV} = [0.05, 0.15]\\
&r^0_{HM} = [0.15, 0.25]\\
&r^{NTI}_{HM} = [0.45, 0.55]\\
&r^0_{PLe} = [0.05, 0.15]\\
&r^{NTI}_{PLe} = [0.15, 0.25]\\
&r^0_{CM} = [0.15, 0.25]\\
&r^N_{CM}TI = [0.35, 0.45]\\
&m_{FT} = [0.0001, 0.0015]\\
&m_{PL} = [0.0001, 0.0015]\\
&m_{DP} = [0.0005, 0.0015]\\
&m_{SV} = [0.0005, 0.0015]\\
&m^0_{HM} = [0.0005, 0.0015]\\
&m^N_{HM}TI = [0.00005, 0.00015]\\
&m_{PLe} = [0.0002, 0.0004]\\
&m_{CM} = [0.00005, 0.00015]
\end{aligned}
$$

In this simulation, only the interactions PL-DP and FT-PL are considered trophic. Therefore, we report only a values for these. The FT-PL interaction is not affected by any third species, but the PL-DP interaction is mediated by the presence of HM plants: a higher abundance of HM flowers increases the probabilities that an interaction takes place, therefore increasing its net outcome.

$$
\begin{aligned}
&a_{FT,PL} = [10^{-6}, 10^{-4}]\\
&a_{PL,FT} = [-10^{-2}, -10^{-4}]\\
&a^0_{PL,DP} = [10^{-7}, 10^{-5}]\\
&a^N_{PL,DP}TI = [10^{-4}, 10^{-2}]\\
&a^0_{DP,PL} = [-10^{-5}, -10^{-7}]\\
&a^N_{DP,PL}TI = [-10^{-2}, -10^{-4}]
\end{aligned}
$$

The N_0 parameters in previous equations represent a typical average abundance of the non-trophic interactor. These values were taken, when possible, from [?, ?]. Diptera densities were approximated based on [?]:

$N^0_{PL} = 2189$
$N^0_{DP} = 150$
$N^0_{SV} = 3000$
$N^0_{HM} = 7187$
$N^0_{PLe} = 187$
$N^0_{CM} = 10000$

Equal Footing Network

The model is of the form:

$$\frac{dN_x}{dt} = r_x N_x \tag{A.9}$$

where

$$r_x = r_x^0 + \sum_{y \in S, y \neq x} a_{xy} N_y - \left(\beta_x + c_x \sum_{y \in S, y \neq x} a_{xy} N_y \right) N_x \tag{A.10}$$

As with the Expanded Food Web model, we constrained each free parameter to a given range and simulated 100000 times the system for 2500 time steps, assigning a random value to each parameter within its range. Here we varied the strength of the antagonistic and facilitative interactions (commensalistic and mutualistic) and checked the stability of the resulting network by means of a local stability analysis.

$r_{FT} = [-0.01, -0.001]$
$r_{PL} = [0.01, 0.1]$
$r_{DP} = [0.4, 0.6]$
$r_{SV} = [0.05, 0.15]$
$r_{HM} = [0.15, 0.25]$
$r_{PLe} = [0.05, 0.15]$
$r_{CM} = [0.15, 0.25]$
$\beta_x = [10^{-5}, 10^{-4}] \forall x$
$c_x = 10^{-3} \forall x$

Interaction strengths
1. Weak interactions

$a_{FT,PL} = [10^{-7}, 10^{-5}]$
$a_{PL,FT} = [-10^{-5}, -10^{-7}]$
$a_{PL,DP} = [10^{-7}, 10^{-5}]$
$a_{PL,HM} = [10^{-7}, 10^{-5}]$
$a_{PL,PLe} = [10^{-7}, 10^{-5}]$
$a_{PL,CM} = [10^{-7}, 10^{-5}]$
$a_{DP,PL} = [-10^{-5}, -10^{-7}]$
$a_{DP,HM} = [10^{-7}, 10^{-5}]$
$a_{HM,PL} = [10^{-7}, 10^{-5}]$
$a_{HM,DP} = [10^{-7}, 10^{-5}]$
$a_{PLe,PL} = [10^{-7}, -10^{-5}]$
$a_{CM,PL} = [10^{-7}, -10^{-5}]$

2. Strong antagonisms

$a_{FT,PL} = [10^{-4}, 10^{-2}]$
$a_{PL,FT} = [-10^{-2}, -10^{-4}]$
$a_{PL,DP} = [10^{-4}, 10^{-2}]$
$a_{PL,HM} = [10^{-7}, 10^{-5}]$
$a_{PL,PLe} = [10^{-7}, 10^{-5}]$
$a_{PL,CM} = [10^{-7}, 10^{-5}]$
$a_{DP,PL} = [-10^{-2}, -10^{-4}]$
$a_{DP,HM} = [10^{-7}, 10^{-5}]$
$a_{HM,PL} = [10^{-7}, 10^{-5}]$
$a_{HM,DP} = [10^{-7}, 10^{-5}]$
$a_{PLe,PL} = [10^{-7}, -10^{-5}]$
$a_{CM,PL} = [10^{-7}, -10^{-5}]$

3. Strong facilitation

$$a_{FT,PL} = [10^{-7}, 10^{-5}]$$
$$a_{PL,FT} = [-10^{-5}, -10^{-7}]$$
$$a_{PL,DP} = [10^{-7}, 10^{-5}]$$
$$a_{PL,HM} = [10^{-4}, 10^{-2}]$$
$$a_{PL,PLe} = [10^{-4}, 10^{-2}]$$
$$a_{PL,CM} = [10^{-4}, 10^{-2}]$$
$$a_{DP,PL} = [-10^{-5}, -10^{-7}]$$
$$a_{DP,HM} = [10^{-4}, 10^{-2}]$$
$$a_{HM,PL} = [10^{-4}, 10^{-2}]$$
$$a_{HM,DP} = [10^{-4}, 10^{-2}]$$
$$a_{PLe,PL} = [10^{-4}, -10^{-2}]$$
$$a_{CM,PL} = [10^{-4}, -10^{-2}]$$

List of Acronyms

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- **Chapter 2:**
García-Callejas, D., Molowny-Horas, R., & Araújo, M. B. (2018). Multiple interactions networks: towards more realistic descriptions of the web of life. *Oikos*, 127, 5-22. doi: 10.1111/oik.04428

- **Chapter 3:**
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- **Chapter 4:**
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- **Chapter 5:**
García-Callejas, D., Gravel, D., Molowny-Horas, R., & Araújo, M. B. Manuscript in preparation.

- **Chapter 6:**
García-Callejas, D., Molowny-Horas, R., & Araújo, M. B. Manuscript in preparation.

N.B.: the code used to generate the results of the different chapters is on the process of being cleaned up, and will be uploaded in its entirety to a public repository, at https://github.com/DavidGarciaCallejas/. As of 20/09/2018, most of the functions related to the model of chapter 3, and the full R code for replicating the results of chapters 4, 5, and 6, are already available. Likewise, the latex files for generating this document are also available. The illustrations for the heading pages of the chapters are all in the public domain, and the authors/name of the works are:

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- Chapter 2: El Perro, by Francisco de Goya
- Chapter 3: Street art by El Niño de las Pinturas
- Chapter 4: Street art by El Niño de las Pinturas
- Chapter 5: Crow and the Moon, by Kawanabe Kyōsai
- Chapter 6: Photograph of Cerro Huenes, Granada, taken by the author
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